

JEE MAIN REVISION MASTER: QUADRATIC EQUATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Roots of Quadratic & Vieta's Relations	2 PYQs	Quadratic Roots
Standard	Location of Roots & Quadratic Inequalities	1 PYQs	Location of Roots

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Roots of Quadratic & Vieta's Relations	$\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$, $ \alpha - \beta = \frac{\sqrt{D}}{ a }$, $aS_n + bS_{n-1} + cS_{n-2} = 0$ (Newton)
Location of Roots & Quadratic Inequalities	$\text{Both roots } > k \iff D \geq 0, -\frac{b}{2a} > k, a \cdot f(k) > 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Roots of Quadratic & Vieta's Relations (Standard Yield — 2 PYQs)

Core Principle: Discriminant analysis for real/complex roots, symmetric root functions, and Newton's power sum theorem.

■ **Caution / JEE Pitfall:** If coefficients a, b, c are rational and D is not a perfect square, roots occur in irrational conjugate pairs $(p \pm \sqrt{q})$.

■ **Speed Trick:** Newton's theorem: If α, β are roots of $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, then $aS_n + bS_{n-1} + cS_{n-2} = 0$.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "If α and β are roots of the equation $x^2 = x + 1 = 0$, then $\alpha^2 + \beta^2$ is..." [Ans: N/A]

3.2 Location of Roots & Quadratic Inequalities (Standard Yield — 1 PYQs)

Core Principle: Conditions for roots relative to real numbers k , k_1 , k_2 , and signs of quadratic expression $ax^2 + bx + c$.

■ **Caution / JEE Pitfall:** For both roots to be greater than k , checking $a \cdot f(k) > 0$ alone is NOT sufficient; you must ALSO check $D \geq 0$ and $-(b/2a) > k$.

■ **Speed Trick:** If $a > 0$, the quadratic $f(x) = ax^2 + bx + c > 0$ for all real x if and only if $D < 0$.

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "Given that a, b and x are real numbers and $a < b, x < 0$, then..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is

- (A) $x^2 - 10x - 16 = 0$
- (B) $x^2 + 10x + 16 = 0$
- (C) $x^2 + 10x - 16 = 0$
- (D) $x^2 - 10x + 16 = 0$

Correct Answer: None

Solution Analysis: Let a and b be the roots the quadratic equation. Then, the quadratic equation is $x^2 - (a+b)x + ab = 0$ (i)
It is given that $AM=5$ and $GM=4$ i.e. $(a+b)/2=5$ and $\sqrt{ab}=4 \Rightarrow a+b=10$ and $ab=16$ On substituting these values in Eq. (i), we obtain. Hence, $x^2 - 10x + 16 = 0$ is the required equation.

Benchmark Problem 2: KCET 2023 (Q#2)

Given that a, b and x are real numbers and $a < b, x < 0$, then

- (A) $(a/x) \geq (b/x)$
- (B) $(a/x) < (b/x)$
- (C) $(a/x) \leq (b/x)$
- (D) $(a/x) > (b/x)$

Correct Answer: None

Solution Analysis: Given, $a < b$ Dividing the above in equation by " x " $\Rightarrow (a/x) > (b/x)$ [■ $x < 0$] [Sign of inequality changes when its both sides are divided by a negative number]